Research Square

Preprints are preliminary reports that have not undergone peer review. They should not be considered conclusive, used to inform clinical practice, or referenced by the media as validated information.

Prediction of 30-day, 90-day and 1 year mortality after colorectal cancer surgery using a data-driven approach

Karoline Bendix Bräuner

karob@regionsjaelland.dk

Zealand University Hospital

Andi Tsouchnika

Zealand University Hospital

Maliha Mashkoor

Zealand University Hospital

Ross Williams

Erasmus MC

Andreas Weinberger Rosen

Zealand University Hospital

Morten Frederik Schlaikjaer Hartwig

Zealand University Hospital

Mustafa Bulut

Zealand University Hospital

Niclas Dohrn

Zealand University Hospital

Peter Rijnbeek

Erasmus MC

Ismail Gögenur

Zealand University Hospital

Research Article

Keywords: colorectal cancer, machine learning, prediction model, mortality, postoperative

Posted Date: November 16th, 2023

DOI: https://doi.org/10.21203/rs.3.rs-3534294/v1

License: This work is licensed under a Creative Commons Attribution 4.0 International License. Read Full License

Additional Declarations: No competing interests reported.

Version of Record: A version of this preprint was published at International Journal of Colorectal Disease on February 29th, 2024. See the published version at https://doi.org/10.1007/s00384-024-04607-w.

Abstract

Purpose: To develop prediction models for short-term mortality risk assessment following colorectal cancer surgery.

Background: Individual risk prediction for mortality following CRC surgery can assist surgeons in clinical decision-making and patient-centered care. Several risk scoring systems exist to assess the risk of postoperative mortality following a CRC resection, but they are based on intra- or postoperative variables or do not represent contemporary clinical practice.

Methods: Data was harmonized from four Danish observational health databases into the Observational Medical Outcomes Partnership Common Data Model. With a data-driven approach using the Least Absolute Shrinkage and Selection Operator logistic regression on preoperative data, we developed 30-day, 90-day, and 1-year mortality prediction models. We assessed discriminative performance using the area under the receiver operating characteristic and precision-recall curve and calibration using calibration slope, intercept, and calibration-in-the-large. We additionally assessed model performance in subgroups of curative, palliative, elective, and emergency surgery.

Results: A total of 57,558 patients were included in the study population. The model showed good discrimination with an area under the receiver-operating characteristic curve of 0.88, 0.878, and 0.861 for 30-day, 90-day, and 1-year mortality respectively, and a calibration-in-the-large of 1.01, 0.99, and 0.99.

Conclusion: We were able to train prediction models for the risk of short-term mortality on a data set of four combined national health databases with good discrimination and calibration.

Introduction

Colorectal cancer (CRC) is the second leading cause of cancer-related death and the third most common malignant neoplastic disease worldwide with an annual incidence of 1.8 million new cases [1, 2]. The cornerstone of curative treatment of CRC is surgery, which is known to have a risk of postoperative complications of up to 47.4 %, 1.4 % maor complications, and mortality of 3.3 % inthe postoperative period [3]. The past decades have seen vast progress in reducing mortality after colorectal cancer surgery [4], treatment and complication rates may improve further by personalizing treatment based on each patient’s individual challenges.

Approximately one third of patients diagnosed with CRC can be considered frail due to age, comorbidity, functional capacity, and lifestyle factors [5, 6], and these patients face a higher risk of post-operative mortality [7]. Several studies suggest that frailty is an independent factor of increased mortality and complications after surgery and that identifying frail patients and optimizing their trajectory in relation to surgery can reduce the risk of postoperative adverse outcomes [5, 8, 9]. Similarly, to the variation in frailty, there is large heterogeneity in terms of age, comorbidities, nutritional deficiencies, and histopathological tumor variation between patients with CRC [3, 10, 11]. Several interventions exist to target these challenges such as iron infusions for anemic patients [12, 13], prehabilitation [14–16], medical nutrition therapy [17], optimization of medication, and even different neoadjuvant treatment strategies depending on the tumor are under development [18]. In order to individualise a patients' treatment based on their risk profile, a clinician requires knowledge of the personalised risk of this patient. Therefore, a tool that can provide these estimates would aid in the process of treatment planning.

With this study, we aimed to develop prediction models using a data-driven approach on preoperative data from multiple nationwide health databases focusing on mortality within 30-days, 90-days and 1-year after CRC surgery, which in the future could be used for patient stratification and personalized treatment.

Methods

Data sources

Data were retrieved from four Danish observational health databases with nationwide coverage. These are: The Danish Colorectal Cancer Group database (DCCG) including information about the patient's CRC history with a 95–99% coverage from 2001–2019 [19, 20], The Danish National Patient Register (DNPR) with trajectory data from all contacts with the secondary healthcare sector from 1976–2019 [21], Register of Laboratory Results for Research (RLRR) containing biochemical and microbiological laboratory results from 2013–2019 [22], and the National Prescription Registry (NPR) containing information about claimed drug prescriptions from 1994–2019 [23] were used.

Access to Danish observational health databases for research does not require ethical approval. This study was approved for Region Zealand under the record number REG-102-2020[24].

Study population and outcomes

We included patients above 18 years of age with a CRC diagnosis and date of surgery between May 1st 2001 to December 31st 2019 in both DCCG and DNPR and where the surgery date in DCCG is matched by the date in DNPR. The outcome of interest was all-cause mortality within 30 and 90 days and 1 year after CRC surgery.

Statistical analyses

Each data source was transformed into the Observational Medicine Outcomes Network Common Data Model (OMOP-CDM) v5.3 and subsequently merged [25]. The OMOP-CDM structure enables the use of open-source analysis tools provided by the Observational Health Data Science and Informatics community (OHDSI) [26].

The initial index date was set to the date of surgery, and times-at-risk were set to 30-day, 90-day, and 1-year in order to assess different parts of the patient trajectory with varying impact of the surgical treatment on the individual survival points covering both immediate and more prolonged impact of surgery.

Data was divided into a training set consisting of a random 75% of patients for model development and 25% of patients were used for internal validation. Three-fold cross-validation was used for hyperparameter optimization. When considering the possibility for missing data, OMOP-CDM requires sex and age, which is present for all records. However, for all other fields, there may be missing data. In DCCG some covariates are mutually exclusive and thus can be separated from missing values such as deficient mismatch repair (dMMR) and proficient mismatch repair (pMMR). In the remaining data sources, however, distinguishing between no record or missing record is not possible due to the nature of data capture from EHR and longitudinal data collection, for instance if there was no record of a specific diagnosis, it was not present in the specific patient. Categorical variables such as procedures and diagnoses were stored by one-hot encoding, causing both negative values and missing data for categorical variables to be interpreted as zero by the model. Missing continuous values were interpreted as NA and as such the record of a missing continuous variable would not be taken into consideration [25, 27, 28]. Apart from age, continuous variables were almost exclusively biochemical results, especially from blood samples.

We trained a Least Absolute Shrinkage and Selection Operator (LASSO) Logistic Regression model for 30-day, 90-day, and 1-year postoperative mortality [29–33]. The LASSO Logistic Regression generally performs well with rare outcomes or sparse data. The advantage of LASSO Logistic Regression is that the model is provided with all covariates above a certain amount, which we predefined as 0.1 % of rcords, from the source data and automatically

selects covariates, which are associated with the outcome and thus shrinks tens of thousands of covariates to only a fraction of relevant covariates. Performance of models was assessed using area under the receiver operating characteristic curve (AUROC) and area under precision-recall curve (AUPRC) for assessment of discrimination including calculating the ratio between incidence and AUPRC, and calibration intercept, calibration slope, and calibration-in-the-large for assessment of calibration. Visual representation of receiver operating characteristic, precision-recall curves, and calibration plots are reported and assessed visually [34].

In order to assess the performance of the model in different clinical scenarios related to acuteness and intent of surgery, we performed subgroup analysis by testing the model on subsets of patients, who underwent curative, palliative, emergency, and elective surgery respectively. In the subgroup analyses the model that was trained on the full training set and subsequently validated on the different subsets of patients.

The tool used for covariate selection and model settings was ATLAS version 2.9.0. For model training R v. 4.0.3 was used with the 'PatientLevelPrediction' package v. 4.3.7. Reporting of outcomes of this study adheres to the Transparent Reporting of a multivariable prediction model for Individual Prognosis Or Diagnosis (TRIPOD) guidelines [35].

Results

Participants

A total of 65,612 patients underwent surgery for CRC from 2001–2019 in DCCG. Of these patients 57,558 patients were included in the study cohort. This cohort consisted of 48.9% female patients, with a median age of 72 years. The remaining patient characteristics are found in Table 1 and boxplots showing the predicted risk in the mortality and no mortality groups are found in Fig. 1A, 2A, and 3A. Incidences of mortality were 4.48%, 6.64%, and 12.8% for 30-day, 90-day, and 1-year mortality respectively.

Table 1
Demographic and clinical parameters for patients included in the cohort

Characteristic	Count (n = 57,558)	%
Number of events		
30-day mortality incidence	2,579	4.48
90-day mortality incidence	3,824	6.65
1-year mortality incidence	7,383	12.8
Basic information		
Male	30,668	53.3
Female	26,890	46.7
Age (median, IQR)	Median	IQR*
	71	64–75
Lifestyle and comorbidity		
ASA* Score	12,682	22.0
1	30,267	52.6
2	13,018	22.6
≥ 3	1,591	2.8
Missing		
WHO* Performance status	13,545	23.5
0	4,943	8.6
1	1,972	3.4
≥ 2	37,098	64.5
Missing		
Charlson's Comorbidity Index	36,749	63.8
0	9,632	16.7
1	5,743	10
2	5,432	9.4
≥ 3	< 6	0
Missing		

**IQR = inter quartile range, ASA = American Society of Anesthesiology, WHO = World Health Organization, T-stage = Tumor size stage, N-stage = lymph node stage, M-stage = remote metastasis stage*

Characteristic	Count (n = 57,558)	%
Number of events		
Alcohol consumption per week	10,499	18.2
0 units	26,951	46.8
1–14 units	3,528	6.1
15–21 units	3,249	5.6
≥ 22 units	13,331	23.2
Missing		
Body Mass Index	1,581	2.7
≤ 18.5	21,283	37.0
18.5–25	17,216	29.9
25–30	5,733	10.0
30–35	1,942	3.4
≥ 35	9,803	17.0
Missing		
Smoking status	9,485	16.5
Smoker	19,367	33.6
Previous smoker	17,531	30.5
Never smoked	11,175	19.4
Missing		
Cancer topography		
Colon cancer	40,160	69.8
Rectum cancer	19,321	33.6
Tumor specific details		

**IQR = inter quartile range, ASA = American Society of Anesthesiology, WHO = World Health Organization, T-stage = Tumor size stage, N-stage = lymph node stage, M-stage = remote metastasis stage*

Characteristic	Count (n = 57,558)	%
Number of events		
T stage*	451	0.8
T0	9,482	16.5
T1	10,256	17.8
T2	27,110	47.1
T3	8,703	15.1
T4	1,556	2.7
Missing		
N stage*	27,789	48.3
N0	12,514	21.7
N1	8,700	15.1
N2	389	0.7
N3	8,166	14.2
Missing		
M stage*	51,864	90.1
M0	8,022	13.9
M1		
Treatment details		
Family history of malignant neoplasm of gastrointestinal tract	14,600	25.4

**IQR = inter quartile range, ASA = American Society of Anesthesiology, WHO = World Health Organization, T-stage = Tumor size stage, N-stage = lymph node stage, M-stage = remote metastasis stage*

Characteristic	Count (n = 57,558)	%
Number of events		
Treatment intent	45,135	78.4
Curative	2,237	3.9
Palliative	406	0.7
Compromised resection	9,780	17.0
Missing		

**IQR = inter quartile range, ASA = American Society of Anesthesiology, WHO = World Health Organization, T-stage = Tumor size stage, N-stage = lymph node stage, M-stage = remote metastasis stage*

Model performance

The models contained 419, 561, and 581 covariates for 30-day, 90-day, and 1-year mortality, respectively. The models with all details can be viewed in supplementary table 1.

The models showed discrimination of AUROC of 0.88, 0.878 and 0.861 (Table 2, Fig. 1B, 2B, and 3B) and AUPRC of 0.353, 0.431, and 0.559 for 30-day, 90-day, and 1-year mortality respectively, where the incidence was 0.448, 0.665, and 0.128 (Table 2, Fig. 1C, 2C, and 3C). In terms of calibration, the models had calibration slopes of 0.96, 1.02, and 1.04, calibration intercepts of 0.02, 0.01, and 0.01, and calibration-in-the-large of 1.01, 0.99, and 0.99 for 30-day, 90-day, and 1-year mortality (Table 2, Fig. 1D, 2D, and 3D). Additionally, the visual assessment of performance in terms of the smooth calibration plots was excellent for 30-, 90-day, and 1-year mortality as can be viewed in Figs. 1D, 2D, and 3D.

Table 2
Cohort, calibration, and discrimination metrics for 30-, 90- and 180-day post-operative mortality using LASSO Logistic Regression

	Number of patients	Number of outcomes	Candidate covariates (included)	AUROC	AUPRC	Calibration in the large	Calibration-slope	Calibration-intercept
30-day mortality	57,558	2,579 (4.48%)	24744 (419)	0.88	0.353	1.01	0.96	0.02
90-day mortality	57,558	3,824 (6.65%)	24744 (561)	0.878	0.431	0.99	1.02	0.01
1-year mortality	57,558	7,383 (12.8%)	24744 (581)	0.861	0.559	0.99	1.04	0.01

Area under the receiver operating characteristic (AUROC), Area under precision recall curve (AUPRC)

In subgroup analysis, the results from the general model including all patients operated for CRC were compared to smaller subgroups of patients undergoing elective surgery (n = 30,167), patients undergoing emergency surgery (n = 4,279), patients undergoing palliative surgery (n = 1,829), and patients undergoing curative surgery (n = 24,598). A comparison of all performance evaluation measures in each subgroup can be viewed in Table 3.

Table 3. *Performance shown with discrimination and calibration evaluation methods in subgroup analysis*

	Cohort	Number of patients	Incidence of outcomes	AUROC	AUPRC (relative to incidence)	Calibration in the large	Calibration slope	Weak calibration-intercept	Hosmer-Lemeshow intercept
30-day mortality	**All surgeries**	57,558	4.48 %	0.880	0.353 (7.8)	1.01	0.96	0.11	0.00
	Elective	51,695	3.1 %	0.886	0.275 (8.9)	1.00	1.12	0.31	0.00
	Emergency	5,691	17.1 %	0.877	0.608 (3.6)	0.98	1.05	0.14	0.00
	Curative	45,135	3.0 %	0.896	0.329 (11)	1.00	1.10	0.23	0.01
	Palliative	2,251	15.0 %	0.865	0.588 (3.9)	0.95	1.06	0.16	0.00
90-day mortality	**All surgeries**	57,558	6.65 %	0.878	0.431 (6.4)	0.99	1.02	0.07	0.00
	Elective	51,695	4.7 %	0.873	0.341 (7.1)	0.99	1.13	0.30	0.01
	Emergency	5,691	23.6 %	0.864	0.671 (2.8)	1.01	1.13	0.12	0.01
	Curative	45,135	4.5 %	0.879	0.367 (8.3)	1.01	1.11	0.24	0.00
	Palliative	2,251	25.1 %	0.829	0.652 (2.6)	1.03	0.99	0.05	0.01
1-year mortality	**All surgeries**	57,558	12.9 %	0.861	0.559 (4.3)	0.99	1.04	0.06	0.00
	Elective	51,695	10.1 %	0.854	0.469 (4.6)	1.00	1.10	0.16	0.01
	Emergency	5,691	37.8 %	0.852	0.781 (2.1)	1.00	1.13	0.05	0.02
	Curative	45,135	9.0 %	0.850	0.429 (4.8)	0.99	1.07	0.11	0.00
	Palliative	2,251	46.6 %	0.812	0.788 (1.7)	0.97	1.03	0.08	0.01

Area under the receiver operating characteristic (AUROC), Area under precision recall curve (AUPRC)

Discussion

In this study, we developed prediction models using machine learning for data-driven covariate inclusion to predict risk of 30-day, 90-day, and 1-year-mortality following CRC surgery with good discrimination and excellent calibration plots by visual assessment, especially for 1-year mortality (Fig. 3). In the subgroup analysis comparing the general population with subgroups of patients undergoing curative, palliative, emergency, or elective surgery we consistently found good performance comparable to the larger study population in each subgroup, showing that the model can be used in multiple circumstances. The increase in calibration slope, which was seen in some of the subgroups (Table 3) and intercept meant that the subgroup analysis showed tendency to underestimate risk.

Other risk assessment tools currently exist, but are not routinely used in the clinical setting, and tend to perform less impressively on internal validation sets [36–38]. Additionally, most prediction studies tend to mainly report AUROC, but not other performance metrics, especially calibration metrics [39]. In this study, we report a range of graphical and numerical discrimination and calibration measures with full transparency, which also lives up to the reporting guidelines in the TRIPOD statement [35].

Using prediction models for clinical risk assessment is known to have implementation challenges, partly because of suboptimal introduction to the clinicians, which in some cases have been seen to cause lack of trust in the prediction [40, 41]. When interpretability comes into question it is, however, important to note that increasing explainability of the model may reduce the performance and clinical applicability [42]. The data driven covariate inclusion means that some included covariates may seem clinically unrelated to the outcome. The more traditional approach is the pre-selection of variables based upon clinical knowledge and existing known causal relationships. By doing a pre-selection the potential search space for the model is vastly reduced and removes the potential for finding new associations that can be predictive of the outcome. As such this method, whilst providing models with a high explainability, may sacrifice the potential for better performance. This study does not report positive-predictive value, negative-predictive value, sensitivity, and specificity, since they require a threshold for high risk. However, establishment of a high risk is greatly dependent on the patient phenotype and having high or low risk as result of a prediction rather than a number, may lead to the model being used as a deciding factor, and not clinical decision support.

One of the main strengths of this study is the use of high-quality observational health databases with national coverage, high rate of completeness [19–21], high granularity in data, and a combination of differently focused data sources. This yields many patients in the cohort, which provides more data for model training but also reduces selection bias from variation in local practices. The merging of different health databases with a different focus; cancer-specific trajectory data, admission-specific data, prescription medicine data, or laboratory results data; also provides the most comprehensive data coverage related to each patient, which is necessary in order to create full patient phenotypes. An additional strength is the testing of the model in a subgroup analysis of different surgical settings; acute, elective, palliative, and curative, where the model showed equal performance in all subgroups. This additional internal validation step supports that the model provides reliable risk predictions for patients in multiple different settings.

Denmark is known to have observational health databases with high coverage, but other countries such as the Netherlands (for instance Netherlands Cancer Registry provided by Netherlands comprehensive cancer organization (IKNL) [43]) and Norway (for instance Cancer Registry of Norway [44]) show similar datasets and coverage to the Danish databases. This means that models likely can be used in different countries with similar datasets. Additionally, the models are trained to use the provided covariates, meaning that models can be used on incomplete datasets.

The data driven approach meant that all available covariates could potentially be included in the model development. The LASSO logistic regression selects covariates for the model based on their impact on the model and excludes covariates with no effect on the prediction. The use of this data-driven covariate selection often leads to many included covariates, which means that these models utilized 419, 561, and 581 one-hot encoded covariates for prediction, which is infeasible for clinical implementation. A step towards making the number of variables smaller is by grouping them into categories or phenotypes, which will make the model more clinically manageable, a so-called parsimonious model [45, 46]. This would however potentially decrease the performance of the model or may introduce bias, when covariates are selectively grouped. The prospect of parsimonious models is interesting for

the next steps of clinical implementation but will require further research into how the number of covariates can be shrunk with minimal model impact. In addition, external validation with other colorectal cancer research groups with an OMOP-CDM would be an important future step to ensure that models are clinically applicable internationally.

Conclusion

We found that the combination of multiple nationwide databases for patients with CRC allowed for development of high performing prediction models for mortality with great calibration and discrimination. By only including pre-operative covariates, the models are usable in the treatment planning phase and may assist clinicians with optimizing an individualized approach to colorectal cancer treatment.

Declarations

Acknowledgements

The authors thank The Danish Health Data Authority, The Danish Clinical Quality Program, and the Danish Colorectal Cancer Group for access to data. We thank edenceHealth for assistance in transformation of data to OMOP-CDM format. Additionally, we thank the European Health Data Evidence Network (EHDEN) for sparring during the process of data transformation to OMOP-CDM, and the PatientLevelPrediction work group in the OHDSI community for development of open source tools making analysis work for this paper possible.

Author contributions

The study design was developed by Karoline Bendix Bräuner, Andreas Weinberger Rosen, and Ismail Gögenur All authors reviewed the manuscript. Karoline Bendix Bräuner, Maliha Mashkoor, Ross Williams, and Andi Tsouchnika. did the data handling, data analysis, tables, and figures. Karoline Bendix Bräuner, Niclas Dohrn, Morten Frederik Schlaikjær Hartwig, and Ross Williams drafted the manuscript. Andreas Weinberger Rosen, Peter Rijnbeek, Ross Williams, and Ismail Gögenur provided additional expert knowledge and consulting during the writing phase.

Competing interests

All authors have approved the submission of this paper. All authors declare no competing interests.

Funding

This project has received funding from the Innovative Medicines Initiative 2 Joint Undertaking (JU) under grant agreement no 806968. The JU receives support from the European Union’s Horizon 2020 research and innovation programme and EFPIA. Funding was also provided by the Region Zealand and the Danish Ministry of Higher Education and Science.

Additionally, the project received financial support from the Novo Nordisk Foundation Project title: Personalized medicine infrastructure using the open-source OMOP common data model including an Electronic Health Record interface

Grant number: NNF21OC0069821

Data availability

Data consisted of the Danish Colorectal Cancer Group database (DCCG), which was provided by the Regional Clinical Quality Program (RKKP), and the Danish National Patient Register (DNPR), Register of Laboratory Results for Research (RLRR), and the National Prescription Registry (NPR) provided by the Danish Health Data Authorities. Accessibility to data is dependent on approval from these regulatory bodies and appropriate applications.

References

1. Center MM, Jemal A, Smith RA, Ward E. Worldwide variations in colorectal cancer. Dis Colon Rectum 2010;53:1099.
2. World Health Organization. WHO Cancer 2020 2020:https://www.who.int/news-room/fact-sheets/detail/c. https://www.who.int/news-room/fact-sheets/detail/c.
3. Knight SR, Shaw CA, Pius R, Drake TM, Norman L, Ademuyiwa AO, et al. Global variation in postoperative mortality and complications after cancer surgery: a multicentre, prospective cohort study in 82 countries. The Lancet 2021;397:387–97.
4. Iversen LH, Ingeholm P, Gögenur I, Laurberg S. Major reduction in 30-day mortality after elective colorectal cancer surgery: A nationwide population-based study in Denmark 2001-2011. Ann Surg Oncol 2014;21:2267–73.
5. Bojesen RD, Degett TH, Dalton SO, Gögenur I. High World Heath Organization Performance Status Is Associated With Short And Long-term Outcomes After Colorectal Cancer Surgery. Dis Colon Rectum 2021:58–66.
6. Michaud Maturana M, English WJ, Nandakumar M, Li Chen J, Dvorkin L. The impact of frailty on clinical outcomes in colorectal cancer surgery: a systematic literature review. ANZ J Surg 2021;91:2322–9.
7. Ommundsen N, Wyller TB, Nesbakken A, Bakka AO, Jordhøy MS, Skovlund E, et al. Preoperative geriatric assessment and tailored interventions in frail older patients with colorectal cancer: a randomized controlled trial. Colorectal Dis 2018;20:16–25.
8. Pandit V, Khan M, Martinez C, Jehan F, Zeeshan M, Koblinski J, et al. A modified frailty index predicts adverse outcomes among patients with colon cancer undergoing surgical intervention. Am J Surg 2018;216:1090–4.
9. Castellví Valls J, Borrell Brau N, Bernat MJ, Iglesias P, Reig L, Pascual L, et al. Colorectal carcinoma in the frail surgical patient. Implementation of a Work Area focused on the Complex Surgical Patient improves postoperative outcome. Cir Esp 2018;96:155–61.
10. Lees J, Chan A. Polypharmacy in elderly patients with cancer: Clinical implications and management. Lancet Oncol 2011;12:1249–57.
11. Kasi PM, Shahjehan F, Cochuyt JJ, Li Z, Colibaseanu DT, Merchea A. Rising Proportion of Young Individuals With Rectal and Colon Cancer. Clin Colorectal Cancer 2019;18:e87–95.
12. Bojesen RD, Ravn J, Rasmus E, Vogelsang P, Grube C, Lyng J, et al. The dynamic effects of preoperative intravenous iron in anaemic patients undergoing surgery for colorectal cancer 2021:2550–8.
13. Okuyama M, Ikeda K, Shibata T, Tsukahara Y, Kitada M, Shimano T. Preoperative iron supplementation and intraoperative transfusion during colorectal cancer surgery. Surg Today 2005;35:36–40.

14. Bojesen RD, Jørgensen LB, Grube C, Skou ST, Johansen C, Dalton SO, et al. Fit for Surgery — feasibility of short-course multimodal individualized prehabilitation in high-risk frail colon cancer patients prior to surgery. Pilot Feasibility Stud 2022:1–13.
15. Berkel AEM, Bongers BC, Kotte H, Weltevreden P, de Jongh FHC, Eijsvogel MMM, et al. Effects of Community-based Exercise Prehabilitation for Patients Scheduled for Colorectal Surgery With High Risk for Postoperative Complications. Ann Surg 2021;Publish Ah.
16. Barberan-Garcia A, Ubré M, Roca J, Lacy AM, Burgos F, Risco R, et al. Personalised Prehabilitation in High-risk Patients Undergoing Elective Major Abdominal Surgery : A Randomized Blinded Controlled Trial. Ann Surg 2018;267:50–6.
17. Bojesen RD. Effect of modifying high-risk factors and prehabilitation on the outcomes of colorectal cancer surgery : controlled before and after study 2022.
18. Chalabi M, Fanchi LF, Dijkstra KK, Van den Berg JG, Aalbers AG, Sikorska K, et al. Neoadjuvant immunotherapy leads to pathological responses in MMR-proficient and MMR-deficient early-stage colon cancers. Nat Med 2020;26:566–76.
19. Ingeholm P, Gögenur I, Iversen LH. Danish colorectal cancer group database. Clin Epidemiol 2016;8:465–8.
20. Klein MF, Gögenur I, Ingeholm P, Njor SH, Iversen LH, Emmertsen KJ. Validation of the Danish Colorectal Cancer Group (DCCG.dk) database - on behalf of the Danish Colorectal Cancer Group. Colorectal Dis 2020;22:2057–67.
21. Schmidt M, Schmidt SAJ, Sandegaard JL, Ehrenstein V, Pedersen L, Sørensen HT. The Danish National patient registry: A review of content, data quality, and research potential. Clin Epidemiol 2015;7:449–90.
22. Arendt JFH, Hansen AT, Ladefoged SA, Sørensen HT, Pedersen L, Adelborg K. Existing data sources in clinical epidemiology: Laboratory information system databases in Denmark. Clin Epidemiol 2020;12:469–75.
23. Pottegård A, Schmidt SAJ, Wallach-Kildemoes H, Sørensen HT, Hallas J, Schmidt M. Data resource profile: The Danish national prescription registry. Int J Epidemiol 2017;46:798.
24. https://en.nvk.dk/how-to-notify/what-to-notify. National Comitee on Health Research Ethics. What to Notify? 2022.
25. Observational Health Data Sciences and Informatics. The Book of OHDSI 2019:1–470.
26. Hripcsak G, Duke JD, Shah NH, Reich CG, Huser V, Martijn J, et al. Observational Health Data Sciences and Informatics (OHDSI): Opportunities for Observational Researchers 2015:574–8.
27. OHDSI, Reps, J., Schumie, M., Rijnbeek, P., Suchard, M.,Williams R. Patient Level Prediction. GitHub 2022. https://github.com/OHDSI/PatientLevelPrediction.
28. Reps J, Schuemie MJ, Ryan PB, Rijnbeek PR. Building patient-level predictive models 2020:1–45.
29. Vogelsang RP, Bojesen RD, Hoelmich ER, Orhan A, Buzquurz F, Cai L, et al. Prediction of 90-day mortality after surgery for colorectal cancer using standardized nationwide quality-assurance data. BJS Open 2021;5.
30. Williams RD, Markus AF, Yang C, Duarte-Salles T, DuVall SL, Falconer T, et al. Seek COVER: using a disease proxy to rapidly develop and validate a personalized risk calculator for COVID-19 outcomes in an international network. BMC Med Res Methodol 2022;22:1–13.
31. Williams RD, Reps JM, Rijnbeek PR, Ryan PB, Prieto-Alhambra D, Sena AG, et al. 90-Day all-cause mortality can be predicted following a total knee replacement: an international, network study to develop and validate a prediction model. Knee Surgery, Sports Traumatology, Arthroscopy 2021.
32. Bräuner KB, Rosen, Andreas W, Tsouchnika, Adamantia, Walbech, Julie S, Gögenur, Mikail, Lin, Viviane A, Clausen, Johan S R, Gögenur I. Developing prediction models for short-term mortality after surgery for colorectal

cancer using a Danish national quality assurance database. Int J Colorectal Dis 2022:1–15.

33. Lin V, Tsouchnika A, Allakhverdiiev E, Rosen AW, Gögenur M, Clausen JSR, et al. Training prediction models for individual risk assessment of postoperative complications after surgery for colorectal cancer. Tech Coloproctol 2022..
34. Van Calster B, Nieboer D, Vergouwe Y, De Cock B, Pencina MJ, Steyerberg EW, et al. A calibration hierarchy for risk models was defined: from utopia to empirical data. n.d.
35. Collins GS, Reitsma JB, Altman DG, Moons KGM. Transparent Reporting of a multivariable prediction model for Individual Prognosis Or Diagnosis (TRIPOD): The TRIPOD Statement. Ann Intern Med 2015;162:55–63.
36. McMahon KR, Allen KD, Afzali A, Husain S. Predicting Post-operative Complications in Crohn's Disease: an Appraisal of Clinical Scoring Systems and the NSQIP Surgical Risk Calculator. Journal of Gastrointestinal Surgery 2020;24:88–97.
37. Ma M, Liu Y, Gotoh M, Takahashi A, Marubashi S, Seto Y, et al. Validation study of the ACS NSQIP surgical risk calculator for two procedures in Japan. Am J Surg 2021;222:877–81.
38. van der Hulst HC, Dekker JWT, Bastiaannet E, van der Bol JM, van den Bos F, Hamaker ME, et al. Validation of the ACS NSQIP surgical risk calculator in older patients with colorectal cancer undergoing elective surgery. J Geriatr Oncol 2022.
39. Yang C, Kors JA, Ioannou S, John LH, Markus AF, Rekkas A, et al. Trends in the development and validation of patient-level prediction models using electronic health record data: a systematic review 2015.
40. Cowley LE, Farewell DM, Maguire S, Kemp AM. Methodological standards for the development and evaluation of clinical prediction rules: a review of the literature. Diagn Progn Res 2019;3:1–23
41. Scott I, Carter S, Coiera E. Clinician checklist for assessing suitability of machine learning applications in healthcare. BMJ Health Care Inform 2021;28.
42. Ghassemi M, Oakden-Rayner L, Beam AL. The false hope of current approaches to explainable artificial intelligence in health care. Lancet Digit Health 2021;3:e745–50.
43. IKNL. Netherlands Cancer Registry 2023. https://iknl.nl/en/ncr (accessed October 31, 2023).
44. CRN. Cancer Register of Norway 2023. https://www.kreftregisteret.no/en/ (accessed October 31, 2023).
45. Sinha P, Delucchi KL, McAuley DF, O'Kane CM, Matthay MA, Calfee CS. Development and validation of parsimonious algorithms to classify acute respiratory distress syndrome phenotypes: a secondary analysis of randomised controlled trials. Lancet Respir Med 2020;8:247–57.
46. Sandokji I, Yamamoto Y, Biswas A, Arora T, Ugwuowo U, Simonov M, et al. A time-updated, parsimonious model to predict AKI in hospitalized children. Journal of the American Society of Nephrology 2020;31:1348–57.

Figures

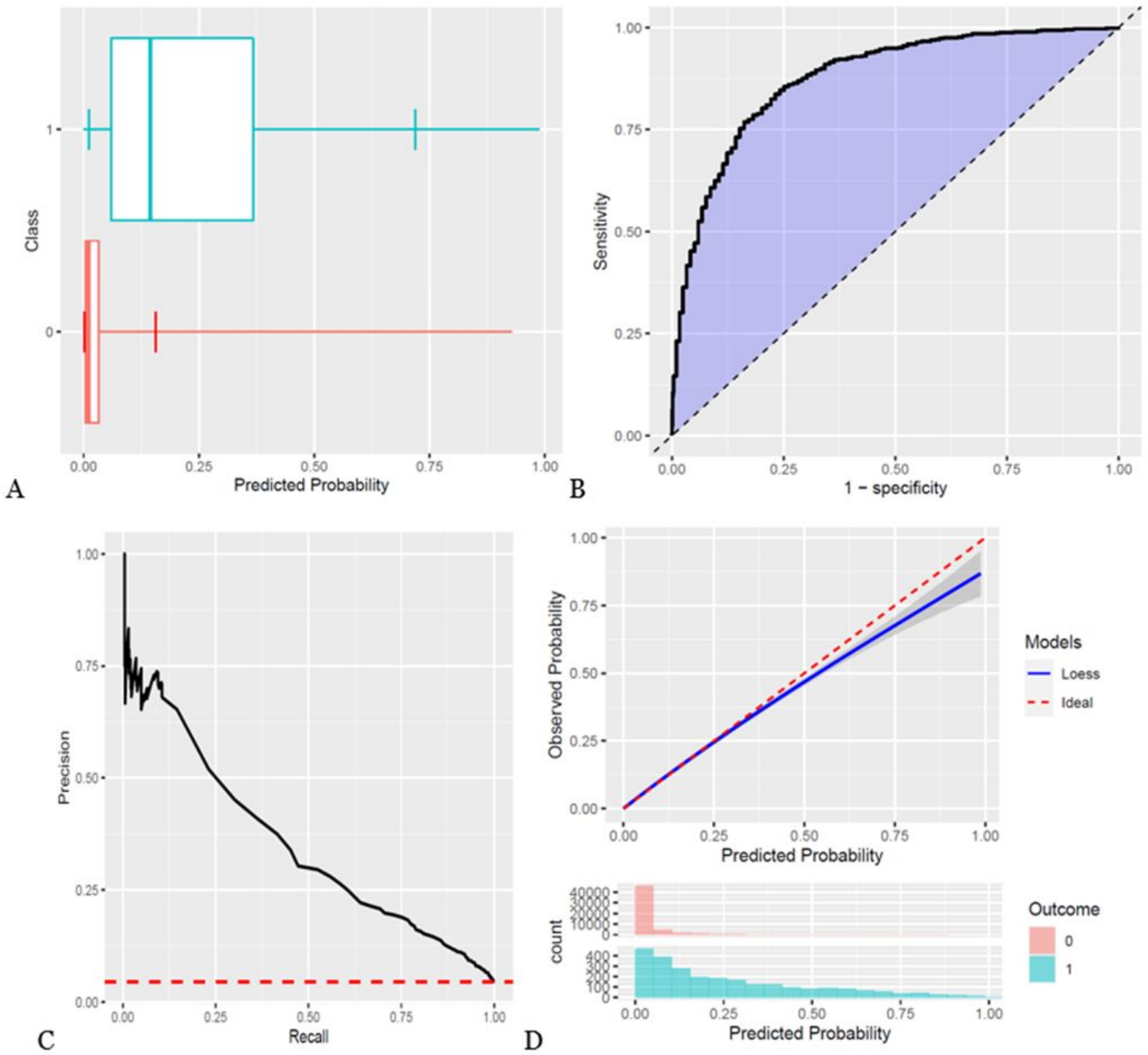

Figure 1

Study outcomes for 30-day mortality. A) Boxplot of predicted risk for patients, who died (Blue) and patients, who did not die (red) B) Receiver operation characteristic (ROC) curve for 30-day mortality C) Precision recall curve for 30-day mortality D) Calibration plot for 30-day mortality

Blue color are cases (mortality within the time at risk), red color is patients, who did not die within the time at risk.

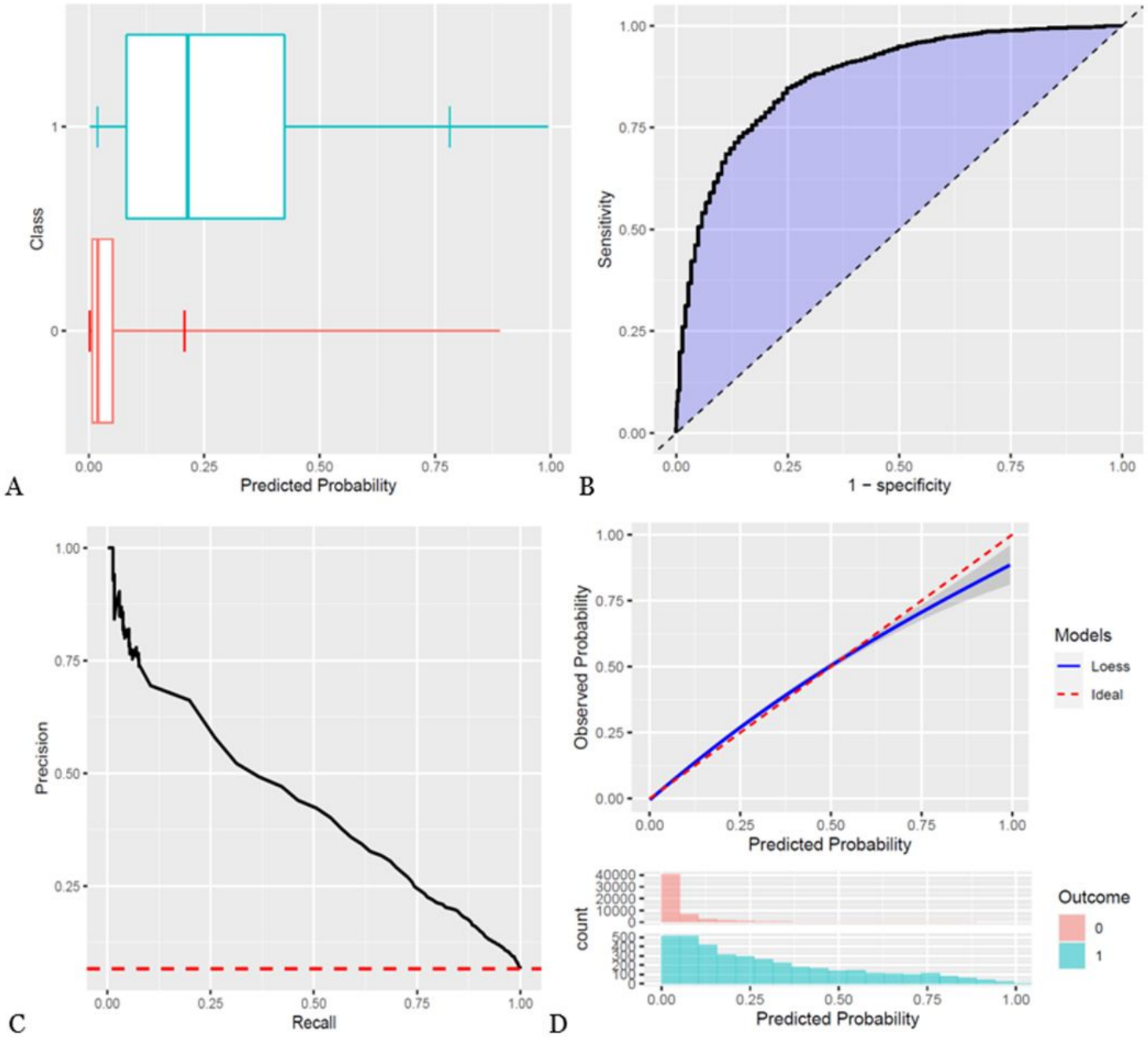

Figure 2

Study outcomes for 90-day mortality. A) Boxplot of predicted risk for patients, who died (Blue) and patients, who did not die (red) B) Receiver operation characteristic (ROC) curve for 90-day mortality C) Precision recall curve for 90-day mortality D) Calibration plot for 90-day mortality

Blue color are cases (mortality within the time at risk), red color is patients, who did not die within the time at risk.

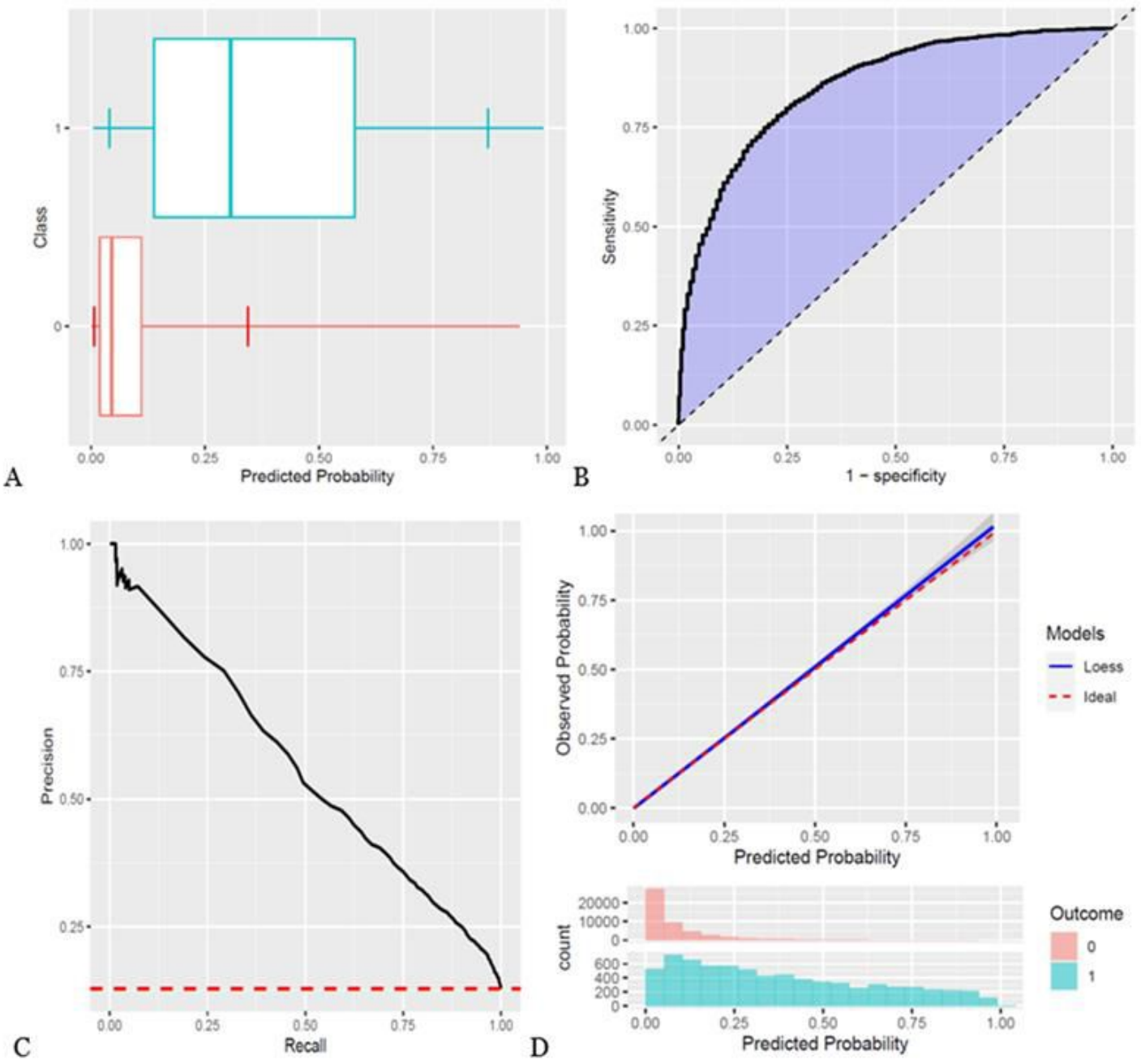

Figure 3

Study outcomes for 1-year mortality. A) Boxplot of predicted risk for patients, who died (Blue) and patients, who did not die (red) B) Receiver operation characteristic (ROC) curve for 1-year mortality C) Precision recall curve for 1-year mortality D) Calibration plot for 1-year mortality

Blue color are cases (mortality within the time at risk), red color is patients, who did not die within the time at risk.

Supplementary Files

This is a list of supplementary files associated with this preprint. Click to download.

- InternationalJournalofColorectalDiseaseSUPPLEMENTARY.docx